MOHAMMAD SAKAWAT KABIR

Security-Focused Backend & Systems Developer | Penetration Testing | DevOps & Cloud Infrastructure

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SUMMARY

Computer Science undergraduate at CUET building, deploying, and securing real-world systems across backend engineering, penetration testing, networking, DevOps, and self-hosted AWS infrastructure. Hands-on experience with Linux, Docker, PostgreSQL, Redis, and GitHub Actions-based CI/CD, applying security principles through authorized labs and self-managed deployments. TryHackMe MAGE tier (Top 4%) and Codeforces Pupil with 500+ problems solved across online judges, bringing a systems-first approach to understanding, testing, and hardening software and infrastructure.

EDUCATION

Bachelor of Science in Computer Science & Engineering

Chittagong University of Engineering & Technology (CUET)

10/2023 - Present Chittagong, Bangladesh

TECHNICAL SKILLS

Programming Languages

Python, TypeScript, C, C++, C#, Java, Bash

Backend & Data Engineering

FastAPI, Django, Spring Boot, PostgreSQL, MySQL, Redis, SQL, ETL, data modeling, data cleaning, data validation

Security & Networking

Penetration testing, network traffic analysis, threat detection, Scapy, eBPF/BCC, DNS, TCP/IP

DevOps, Systems & Infrastructure

Linux, Docker, AWS EC2, Git, GitHub Actions, CI/CD, Nginx Proxy Manager, Twingate

PROJECTS

Real-Time Cyber Attack Monitoring & Threat Intelligence Platform

Tools: TypeScript, WebSockets, Machine Learning, Security Monitoring

- Built and self-hosted a real-time security-monitoring platform that detects SQL injection, XSS, brute-force and DDoS activity through a hybrid rule-based and machine-learning engine.
- Developed a lightweight log agent and WebSocket event pipeline that streams detected activity to a live dashboard for centralized monitoring and investigation.

CinemaSeat — Concurrency-Safe Booking Platform

Tools: FastAPI, React, TypeScript, PostgreSQL, Redis, GitHub Actions, CI/CD

- Built CinemaSeat with team Cuet cryptic_knights, earning **1st Runner-Up** at Zero to Production: The Ultimate Hackathon 2026.
- Combined atomic Redis seat holds using SET NX EX with PostgreSQL uniqueness constraints to make double-booking structurally preventable under concurrent requests.
- Validated the design with a 100-buyer contention test that produced one successful booking, 99 clean conflicts and zero oversells; automated validation and delivery through GitHub Actions CI/CD.

Full Custom DNS Spoofer with Scapy

Tools: Python, Scapy, ARP, DNS, Kali Linux

- Built an educational toolkit covering network discovery, MAC manipulation, ARP spoofing, packet inspection and targeted DNS-response interception.
- Conducted all testing within isolated, authorized lab environments and documented the packet flow, defensive detection signals and relevant mitigations.

PROJECTS

Self-Hosted Cloud Server

Tools: Linux, Docker, Nextcloud, MariaDB, Redis, Pi-hole, Nginx Proxy Manager, Twingate

- Deployed and operated a containerized private-cloud stack providing file storage, synchronization, HTTPS routing, local DNS and secure remote connectivity.
- Integrated Nextcloud, MariaDB, Redis and Pi-hole behind Nginx Proxy Manager, with Twingate providing zero-trust remote access.

Hermes Cloud Agent Lab

Tools: AWS EC2, Linux, Shell, Ollama, Telegram, Gemma GGUF

- Built a reproducible personal AI-agent stack on a low-cost AWS EC2 VPS using Hermes Agent, Ollama and a Telegram gateway.
- Configured model routing between locally hosted Gemma GGUF models and cloud LLMs, documenting deployment and service configuration for repeatability.
- Gained practical experience in Linux service operation, cloud deployment, model hosting and infrastructure troubleshooting.

Basilisk Shell

Tools: Python, Bash, Linux, Natural-Language Processing

- Built an AI-assisted Linux terminal tool that converts natural-language requests into reviewable Bash commands.
- Added layered safety checks, risk-gated execution, multilingual input handling and command-history logging to prevent unsafe commands from being executed blindly.

Spandan — Appointment & Chamber Management Platform

Tools: Python, React, FastAPI, PostgreSQL, Groq

- Developed a full-stack appointment-booking and chamber-management platform with real-time queue management and role-specific workflows.
- Built the backend with FastAPI and PostgreSQL and developed a React interface for patients and healthcare providers.

AgriLink — Role-Based Agricultural Marketplace

Tools: React, TypeScript, Supabase, Row-Level Security, Stripe

- Developed a role-based marketplace connecting farmers, buyers and administrators through dedicated dashboards and workflows.
- Used Supabase and row-level security policies to control data access and integrated Stripe for payment processing.
- Added an AI-assisted chatbot to support platform navigation and user interactions.

CUET Insights — University Community Platforms

Tools: Java, Spring Boot, MySQL, JWT

- Developed a role-aware platform for CUET students, faculty members and alumni, centralizing complaint tracking, innovation showcases and resource sharing.
- Built the backend with Spring Boot and MySQL, using JWT authentication to protect role-specific functionality.

CERTIFICATIONS

Data Engineer Associate
DataCamp

CompTIA Network+ (N10-009) — Course Completion
CBT Nuggets

AWARDS & ACHIEVEMENTS

1st Runner-Up, Zero to Production: The Ultimate Hackathon

Built CinemaSeat with team Cuet cryptic_knights at the IEEE Computer Society CUET × Poridhi.io hackathon.

111th Place, Inter-University Programming Contest, CUET CSE FEST 2025

Competed against university teams from across Bangladesh.

Honorable Mention, ICPC Asia Dhaka Preliminary 2024

Competed in the Online Preliminary Contest as a CUET student.

7th Place, Intra-University Junior Programming Contest, CUET Computer Club

Placed seventh in the intra-university contest organized by the CUET Computer Club.

Pupil Rank, Codeforces

Current rating 1201; maximum rating 1226; 500+ problems solved across online judges.

TryHackMe, MAGE Tier, Top 4%

Completed 102 rooms, earned 12 badges and maintained a 91-day streak.